

AI训练与推理场景下的镜像分发实践

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02 AI镜像分发实践

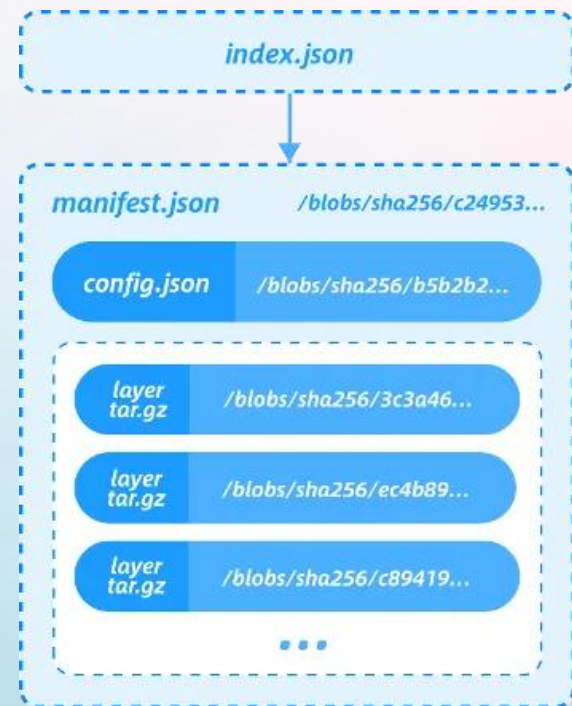
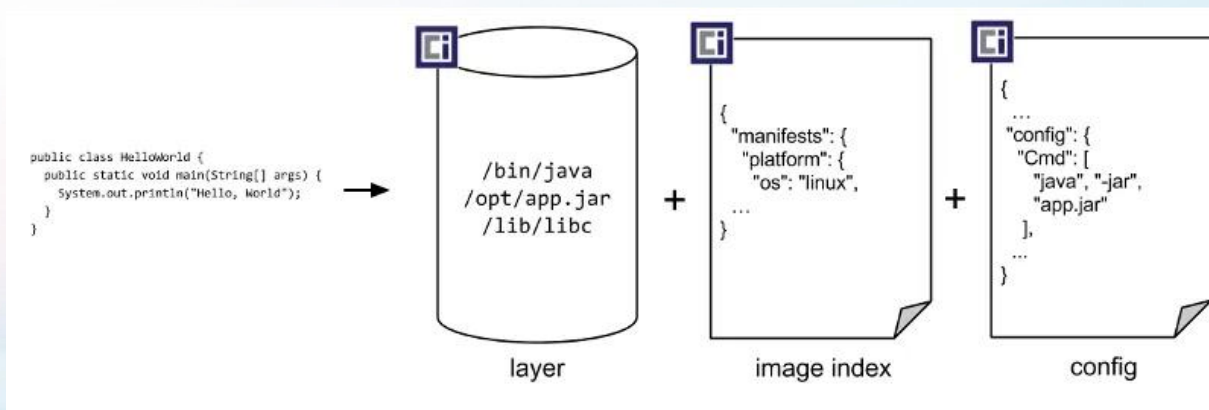
03 模型权重镜像分发实践

04 后续规划

OCI镜像简介

OCI(Open Container Initiatives)Image Spec 定义容器镜像格式

- index
- manifest
- config
- layers



OCI镜像的问题

- **镜像层必须完整下载并解压**

- 镜像拉取时间占容器启动时间的76%
- 只有6.4%数据被容器实际读取

- **元数据更新，导致整个镜像层重新存储和下载**

- **被删除与修改多次的数据，依然会被下载**

- **数据去重率低，层间/镜像间冗余数据**

- **AI相关的镜像越来越大**

- 引擎镜像：数十GB（构建 扫描 拉取耗时都很长）
- 模型镜像：上百GB，TB级别

- **OCI格式的引擎镜像拉取时间，达到分钟级别**

- 业务容器启动耗时长，降低GPU的利用率
- 无法快速扩容满足业务需求

OCI镜像的改进--NyduS



- CNCF毕业项目Dragonfly的镜像加速子项目
- 阿里云，蚂蚁集团，字节跳动等联合开发
- 大规模生产环境验证，蚂蚁每日千万级别容器创建



Production Ready !



Alibaba Cloud

“ Aliyun serverless image pull time drops from 20 seconds to 0.8s seconds. ”



ANT GROUP

“ Serving large-scale clusters with millions of container creations each day. ”



字节跳动
ByteDance

“ Serving container image acceleration in Technical Infrastructure of ByteDance. ”

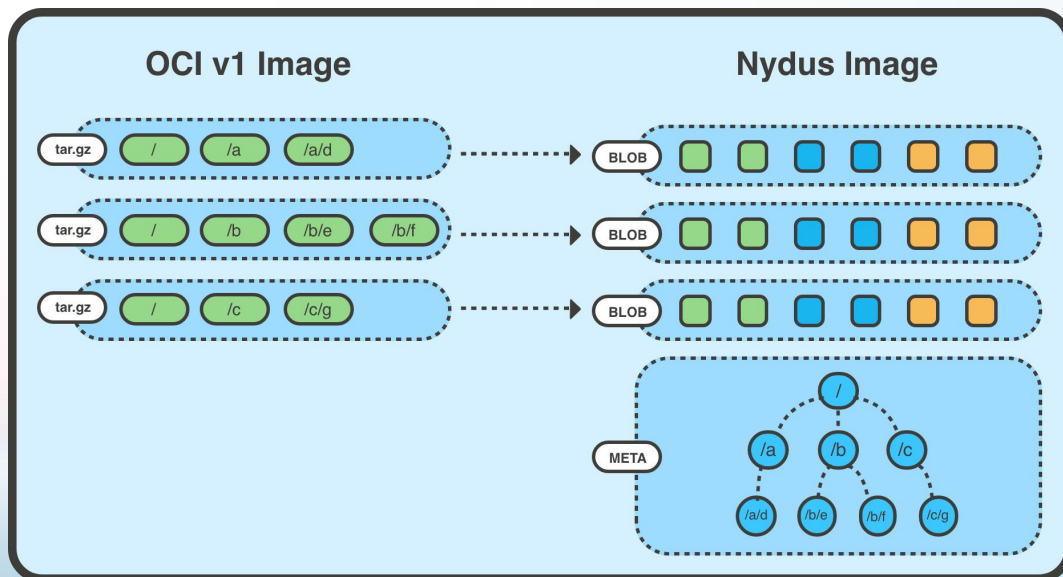
 **NYDUS** | **Acceleration Framework**

OCI镜像的改进--Nydus



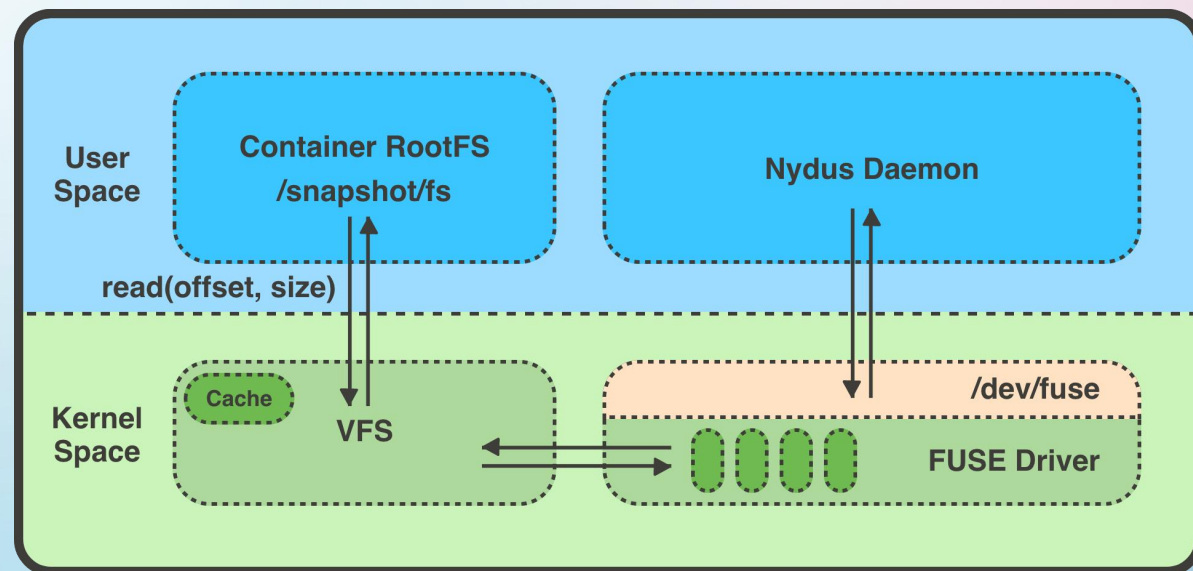
OCI格式改进

- 数据和元数据分离
- Chunk级别组织数据



按需加载

- Fuse 处理文件读IO
- 按需拉取镜像数据



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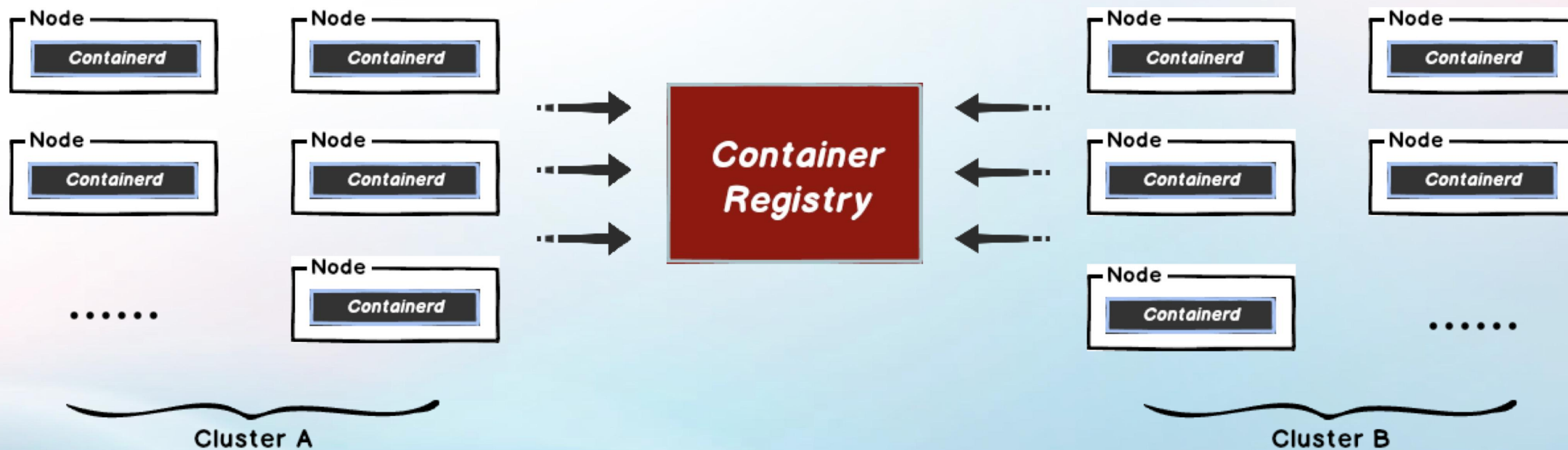
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镜像分发困局

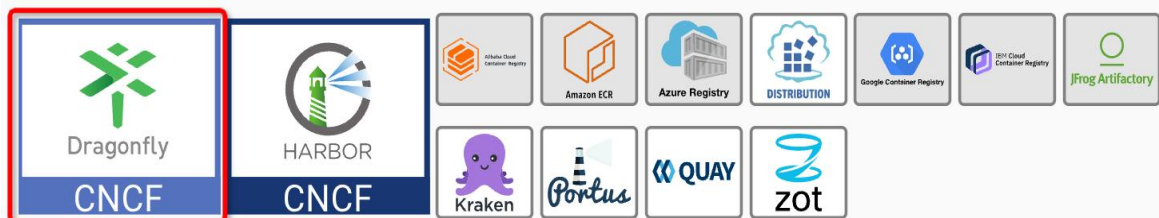
- 源站带宽受限
- 如何利用集群内网带宽



镜像P2P分发加速--Dragonfly



Container Registry



Store & Format



基于 P2P 的镜像加速和文件分发系统

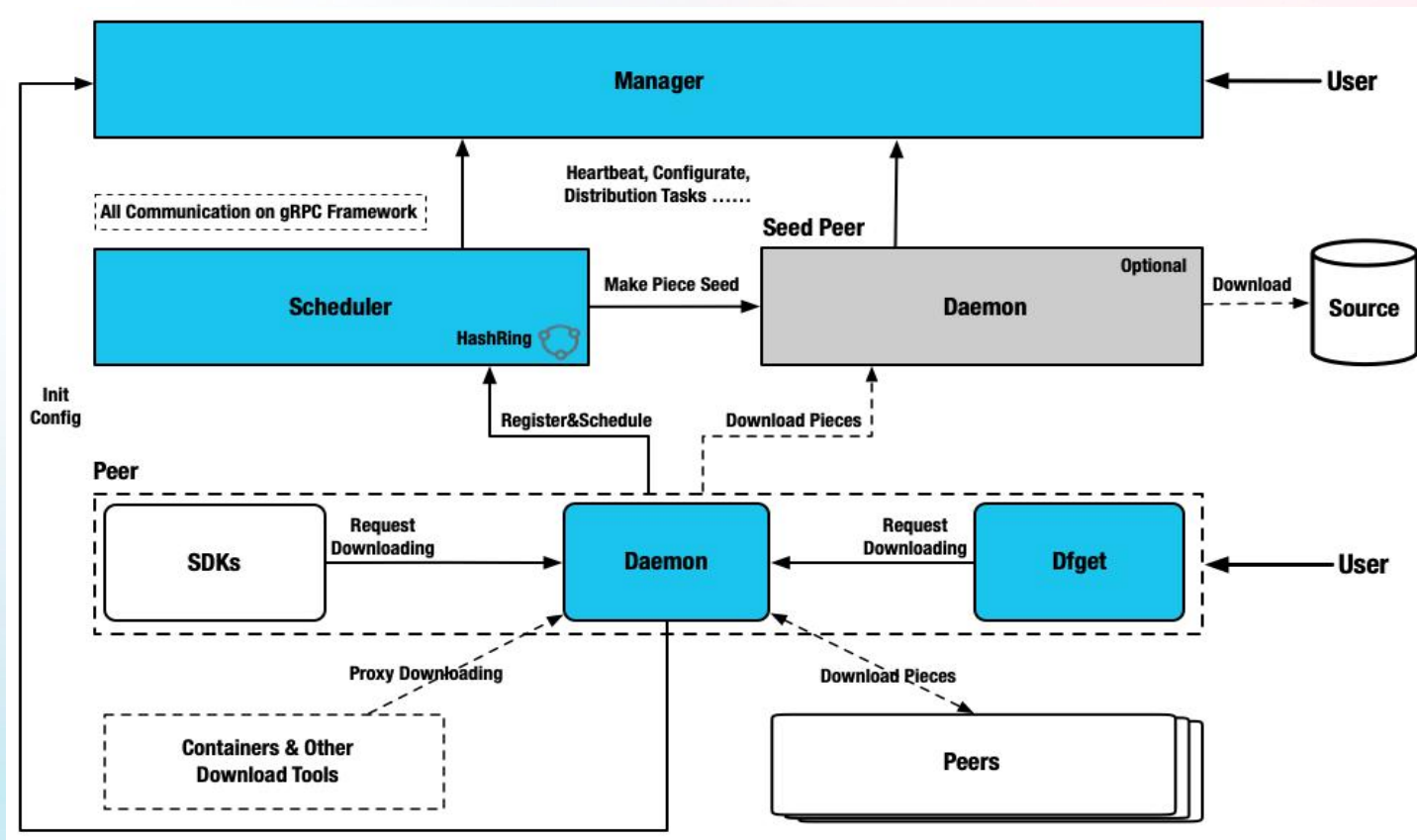
- CNCF毕业项目
- 最大限度的利用节点之间闲置带宽
- 极大减少回源流量

– Nydus小IO优化 降低90%回

源

镜像P2P分发加速--Dragonfly架构 Dragonfly

- **Manager** 管理集群关系、动态配置并提供一个控制台
- **Scheduler** 为下载节点调度一组父节点
- **Peer** 提供了上传和下载功能





镜像加速—Dragonfly+Nydus



Dragonfly

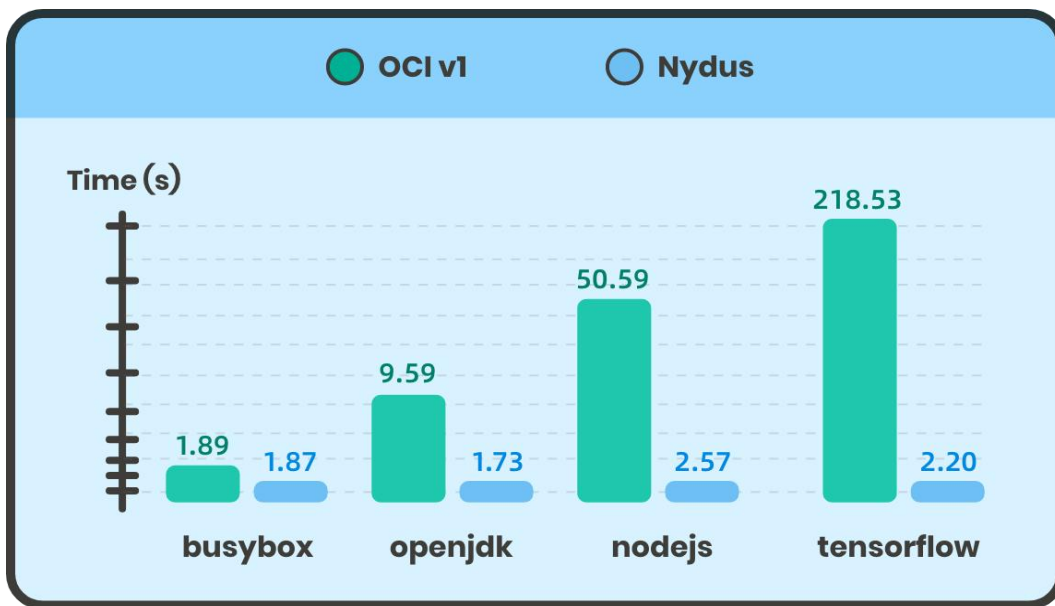


NYDUS

极速启动AI容器



Nydus按需加载实现容器的秒级启动



Pod Creation to Container Start

Sglang推理引擎镜像

- 33GiB中实际读取的数据1.7GiB, 占比5%
- OCI格式启动耗时: 83.2s
- Nydus格式启动耗时: 1.2s

镜像快速构建和扫描



镜像构建支持Build From Nydus

- Base镜像大部分不被使用
- From OCI 16min 6s 16GiB
- From Nydus 1min15s 255MB

镜像安全扫描场景

- 仅扫描部分数据（配置、版本号等）
- 使用Nydus镜像，降低到分钟级别

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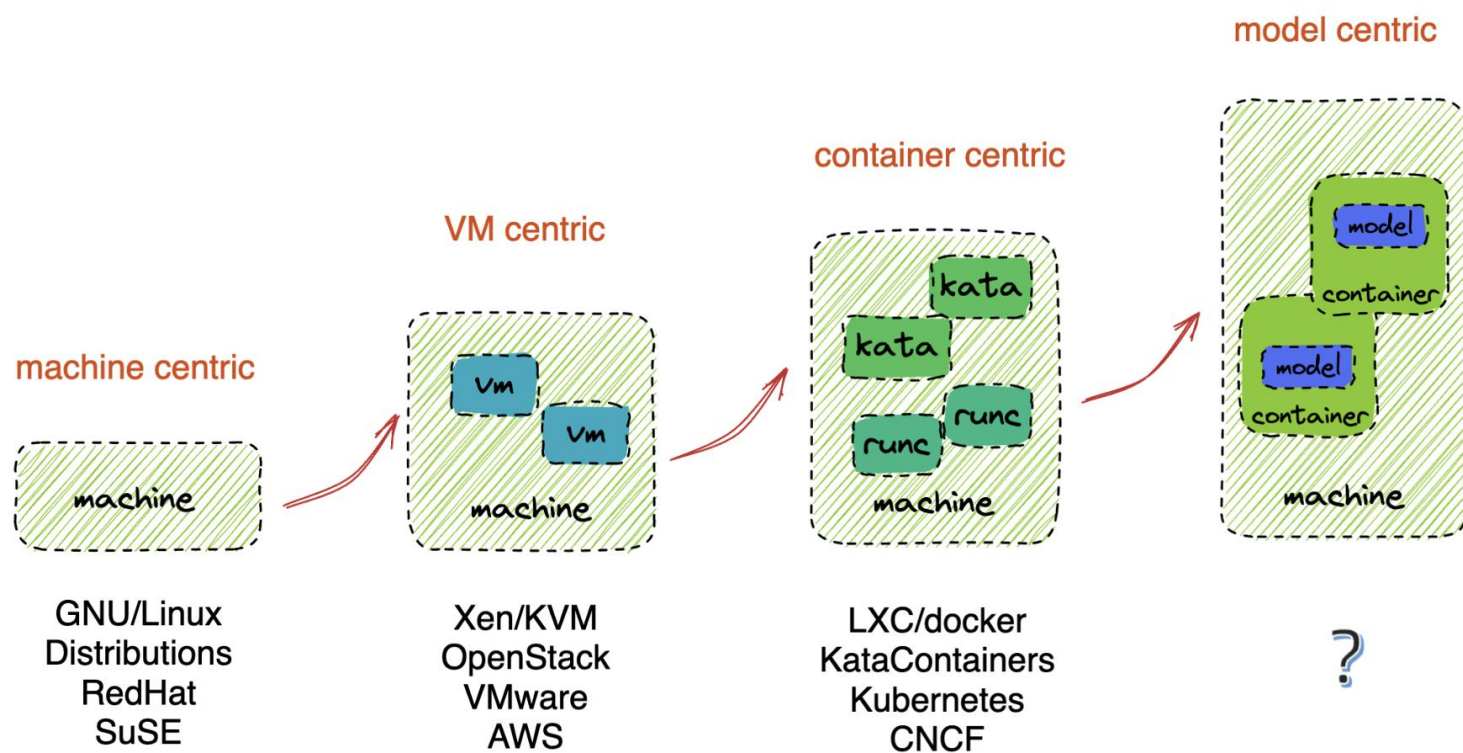
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模型为中心的基础设施



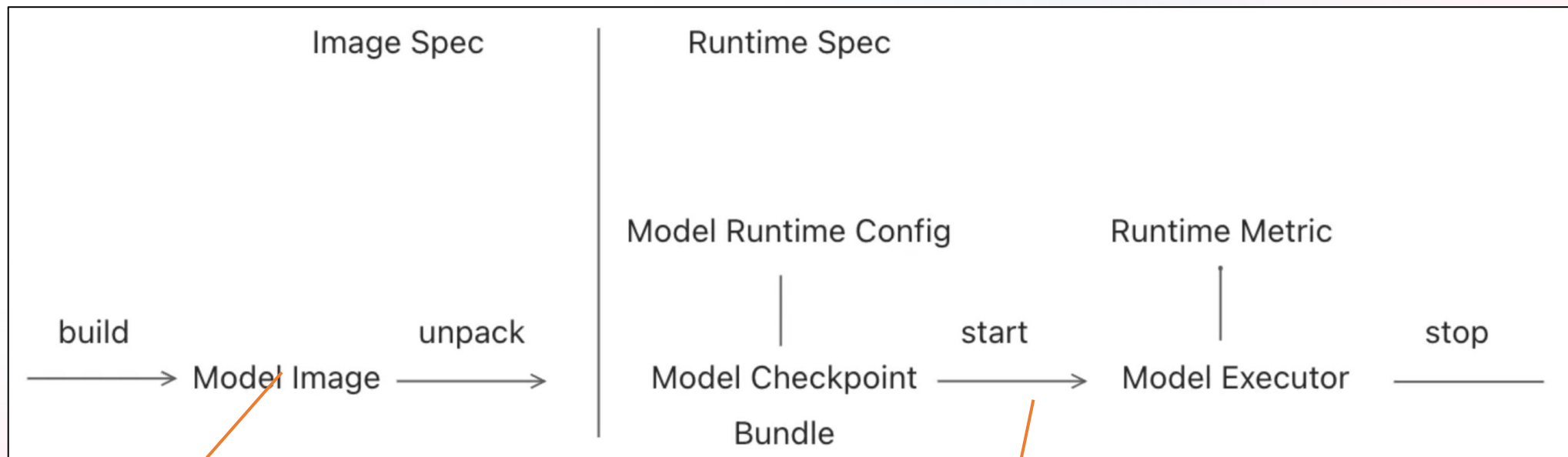
基础设施正在变成以模型为中心

OCI镜像SPEC是软件打包和分发的标准

准
AI模型的打包和分发需要新标准

模型打包分发的开放标准——ModelPack

ModelPack: An Open Standard for Packaging, Distributing, and Running LLMs in Cloud-Native Environments



Model Image Spec

基于OCI Artifact Spec, 定义了AI模型镜像的格式, 包括模型的manifest、layers、config、annotation ...

Model Runtime Spec (WIP)

定义模型的Lifecycle、Runtime Config、Filesystem Bundle、Architecture ...

Model Image Spec

annotations (略)

manifest

```
{
  "schemaVersion": 2,
  "mediaType": "application/vnd.oci.image.manifest.v1+json",
  "artifactType": "application/vnd.cncf.cnai.model.manifest.v1+json",
  "annotations": {
    "org.cnai.model.format.name": "safetensors",
    "org.cnai.model.format.version": "0.4.5",
    ...
  },
  "config": {
    "mediaType": "application/vnd.oci.image.config.v1+json",
    "digest": "sha256:44136fa355b3678all46ad...",
    "size": 2
  },
  "layers": [
    {
      "mediaType": "application/vnd.oci.image.layer.v1.tar",
      "artifactType": "application/vnd.cncf.cnai.model.layer.v1.tar",
      "digest": "sha256:3f4d90098f5b5a6f6a76e9d...",
      "size": 2219940
    },
    {
      "mediaType": "application/vnd.oci.image.layer.v1.tar",
      "artifactType": "application/vnd.cncf.cnai.model.layer.v1.tar",
      "digest": "sha256:760cd441450d65f03710ad...",
      "size": 10865
    },
    {
      "mediaType": "application/vnd.oci.image.layer.v1.tar",
      "artifactType": "application/vnd.cncf.cnai.model.layer.v1.tar",
      "digest": "sha256:10ff7ae0a5c3a62cb7a36f6...",
      "size": 165
    },
    ...
  ]
}
```

config

```
{
  "descriptor": {
    "createdAt": "2025-01-01T00:00:00Z",
    "authors": ["xyz@xyz.com"],
    "vendor": "XYZ Corp.",
    "family": "xyz3",
    "name": "xyz-3-8B-Instruct",
    "version": "3.1",
    "title": "XYZ 3 8B Instruct",
    "description": "xyz is a large language model.",
    "docURL": "https://www.xyz.com/get-started/",
    "sourceURL": "https://github.com/xyz/xyz3",
    "revision": "1234567890",
    "licenses": ["Apache-2.0"]
  },
  "config": {
    "architecture": "transformer",
    "format": "pytorch",
    "paramSize": "8b",
    "precision": "fp16",
    "quantization": "gptq"
  },
  "modelFs": {
    "type": "layers",
    "diff_ids": [
      "sha256:1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef1234567890",
      "sha256:1234567890abcdef1234567890abcdef1234567890abcdef1234567890"
    ]
  }
}
```

layers

• **Layers** array of objects

- **mediaType** string

Implementations SHOULD support the following media types:

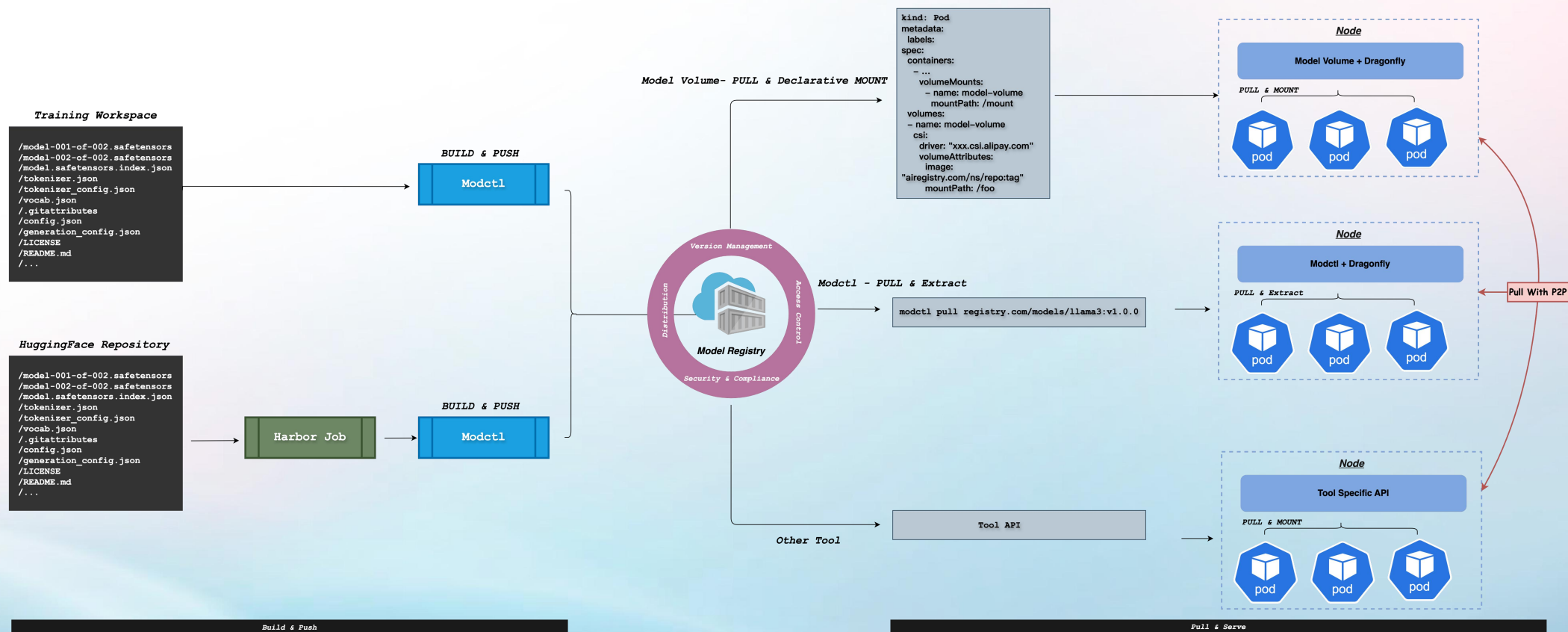
- **application/vnd.cnai.model.weight.v1.raw**: The layer is an unarchived, uncompressed model weights file. If the model weight files are large, implementations are RECOMMENDED to use this media type.
- **application/vnd.cnai.model.weight.v1.tar**: The layer is a [tar archive](#) that contains the model weight file. If the model has multiple weight files, they SHOULD be packaged into separate layers.
- **application/vnd.cnai.model.weight.v1.tar+gzip**: The layer is a [tar archive](#) that includes the configuration file for the model weights. The archive is compressed with [gzip](#).
- **application/vnd.cnai.model.weight.v1.tar+zstd**: The layer is a [tar archive](#) that includes the configuration file for the model weights. The archive is compressed with [zstd](#).
- **application/vnd.cnai.model.weight.config.v1.raw**: The layer is a [tar archive](#) that includes config of the model weights like tokenizer.json, config.json, etc. If the model weight config files are large, implementations are RECOMMENDED to use this media type.
- **application/vnd.cnai.model.weight.config.v1.tar**: The layer is a [tar archive](#) that includes config of the model weights like tokenizer.json, config.json, etc.
- **application/vnd.cnai.model.weight.config.v1.tar+gzip**: The layer is a [tar archive](#) that includes config of the model weights like tokenizer.json, config.json, etc. The archive is compressed with [gzip](#).
- **application/vnd.cnai.model.weight.config.v1.tar+zstd**: The layer is a [tar archive](#) that includes config of the model weights like tokenizer.json, config.json, etc. The archive is compressed with [zstd](#).
- **application/vnd.cnai.model.doc.v1.raw**: The layer is an unarchived, uncompressed documentation file. If the documentation files are large, implementations are RECOMMENDED to use this media type.

定义 raw 类型的media type

- 用于大模型权重文件
- 压缩比低
- 零开销

基于ModelPack的模型管理链路

Modctl: Build、Push、Pull



Model Volume

Kubernetes 1.31:

Read only Volumes Based On OCI Artifacts

```
apiVersion: v1
kind: Pod
metadata:
  name: pod
spec:
  containers:
    - name: test
      image: registry.k8s.io/e2e-test-images/echoserver:2.3
      volumeMounts:
        - name: model
          mountPath: /model
  volumes:
    - name: model
      image:
        reference: harbor.example.com/example/llama:v1.0.0
        pullPolicy: IfNotPresent
```

Model CSI

- 支持把Model Pack模型镜像挂载成Volume
- 简单易用，低版本 K8S 兼容
- 原生支持 Dragonfly 加速

```
apiVersion: v1
kind: Pod
metadata:
  ...
spec:
  ...
  containers:
    - name: main
      volumeMounts:
        # Mount model to /home/admin/model directory
        - name: model-volume-1
          mountPath: /home/admin/model
          readOnly: true
  volumes:
    - name: model-volume-1
      csi:
        driver: "model.csi.alipay.com"
        volumeAttributes:
          # Model format type: image
          "model.csi.alipay.com/type": "image"
          # Model reference url
          "model.csi.alipay.com/reference": "example.com/model/llama:v1.0.0"
```

模型挂载优化

Nydus按需加载模型?

- 所有权重都会被读取
- 存在Fuse开销大
- 压缩去重率较低

Model Volume 直接挂载

- raw格式 避免untar 开销
- 使用hardlink直接复用节点P2P cache

- `layers` *array of objects*

- `mediaType` *string*

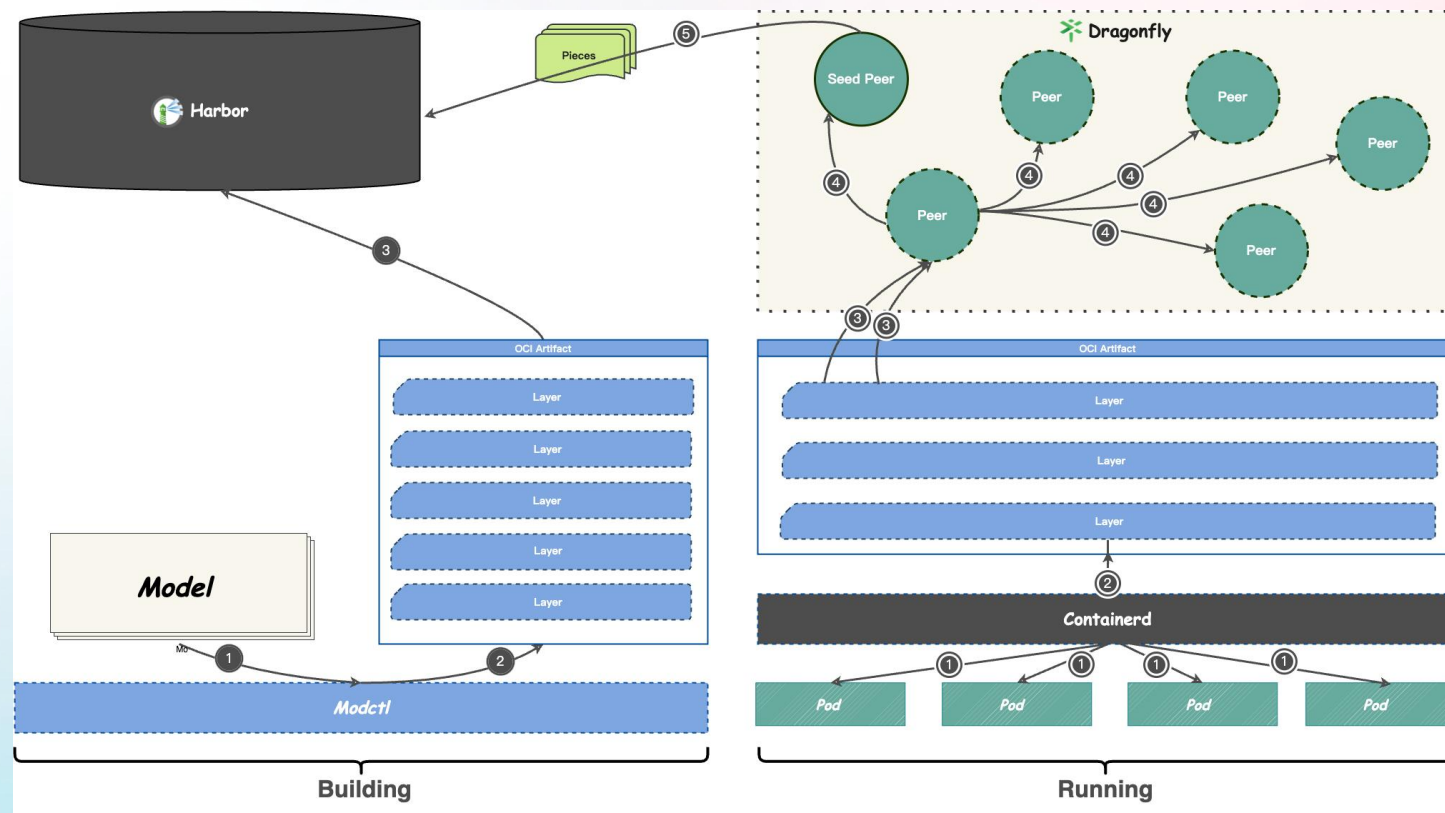
Implementations SHOULD support the following media types:

- `application/vnd.cncf.model.weight.v1.raw` : The layer is an unarchived, uncompressed model weights file.

模型分发加速

使用Dragonfly进行P2P加速分发

- DeepSeek-R1 640GB 分发时间
- 40节点Avg:138s; P95: 155s; P99 198s
- Peer to Peer: 82s



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后续规划

Nydus

- blob布局优化，更好的读放大
- 优先预取AI镜像按需数据

Dragonfly

- P2P调度优化
- RDMA支持
- IO请求分级

ModelPack

- 运行时SPEC支持

Thanks!



Dragonfly



Nydus



ModelPack